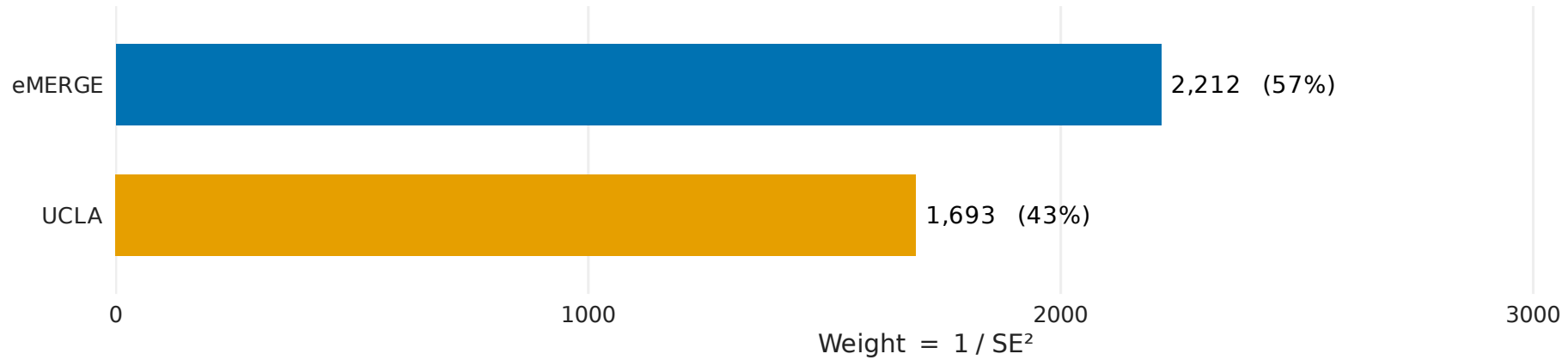


3. Inverse-variance weighting is a balance, not a vote

Weight = $1 / SE^2$. eMERGE has the smaller SE, so it holds about 57% of the weight.

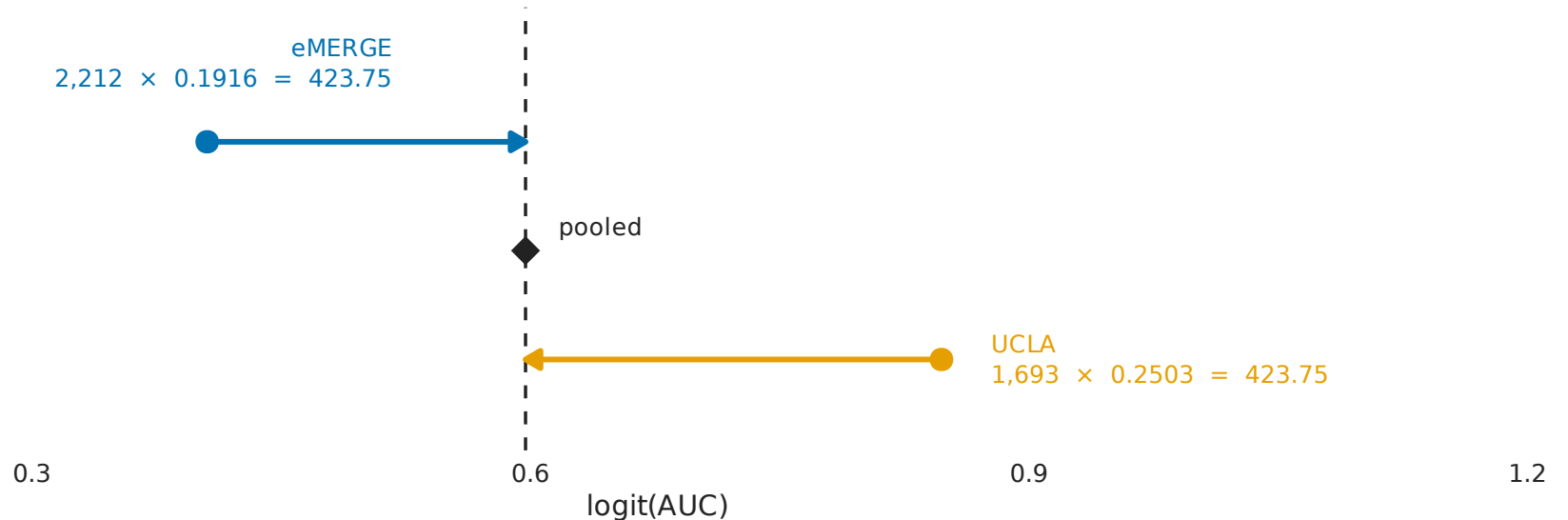
The common value moves until weight times distance is the same on both sides.

Weight of each study



The pooled logit is where the pulls cancel

Both products equal 423.75. The heavier study sits closer to the dashed line.



Pooled logit = $(w1 \times \eta1 + w2 \times \eta2) / (w1 + w2) = 0.597$. SE of that mean = $1 / \sqrt{\text{sum of weights}} = 0.0160$. AUC = $\text{plogis}(0.597) = 0.645$.